Weighted distinct add11 validity

31 October 2026 UTC. Producer field-value proof, pending independent source-fidelity review.

Five Lean declarations compile and pass separate axiom readback. With four nonzero input weights, distinct normalized affine x coordinates and two degree-one Mumford-valid inputs, the weighted source formula produces nonzero output weights and normalized output validity.

Literal field expressions. Put $x_a = a_x/a_z^2$, $y_a = a_y/(a_z^5 a_w)$ and likewise for b. Source cross values are

$$A = a_x b_z^2, \quad B = b_x a_z^2, \quad Y = a_y(b_z^5 b_w), \quad Z = b_y(a_z^5 a_w).$$

The output is

$$q = -(A+B),\ r = AB,\ s = Y - Z,\ t = AZ - BY,\ z = a_z b_z,\ w = (A-B)(a_w b_w).$$

The proof derives $A - B = (x_a - x_b)a_z^2 b_z^2 \neq 0$, then

$$q/z^2 = -(x_a + x_b), \qquad r/z^4 = x_a x_b,$$

$$s/(z^3 w) = (y_a - y_b)/(x_a - x_b),$$

$$t/(z^5 w) = (x_a y_b - x_b y_a)/(x_a - x_b).$$

These identities are conclusions, not premises. The reviewed interpolation/Bezout theorem then yields quadratic support divisibility of $f - v^2$. Cross inequality also excludes inverse and doubling guards in their actual source order.

Controls. Over Q, $a = (0, 0, 2, 3)$ and $b = (9, 486, 3, 2)$ normalize to $(0, 0), (1, 1)$ on $f = X^5 - X + X^2$. The full theorem applies after explicit input divisibility proofs. Checks give $z = 6, w = -216, q/z^2 = -1$ and slope1. Additional controls cover unit weights, zero-z failure and different raw x values encoding equal normalized x.

Receipts and failures. Initial source used reserved keyword "by" as a local name and failed; its diagnostics remain. Corrected coefficient and full-theorem stages, controls, and separate readback exit0. Final axioms are only propext, Classical.choice and Quot.sound; unused-tactic warnings remain recorded. Existing pinned dependencies were used, with120-second per-command limits.

Next machine obligation. The intended native mapping requires odd prime $5 \leq p < 2^{31}$, eight canonical input words, Field.n=1, tables=None and normal execution/counter headroom. Canonical primitive products are below 2^{62} and add/sub intermediates below 2^{32}. The actual UInt64 pow5 and complete expression/guard correspondence is not proved here. Neither are native validity refinement, counters/compiler behavior, divisor-class addition or whole-walk closure.

Source hashes, exact hypotheses and receipts: sources.json, status.json, PLAN.md, attempts and readback1/. No scientific run, root edit or cost measurement.